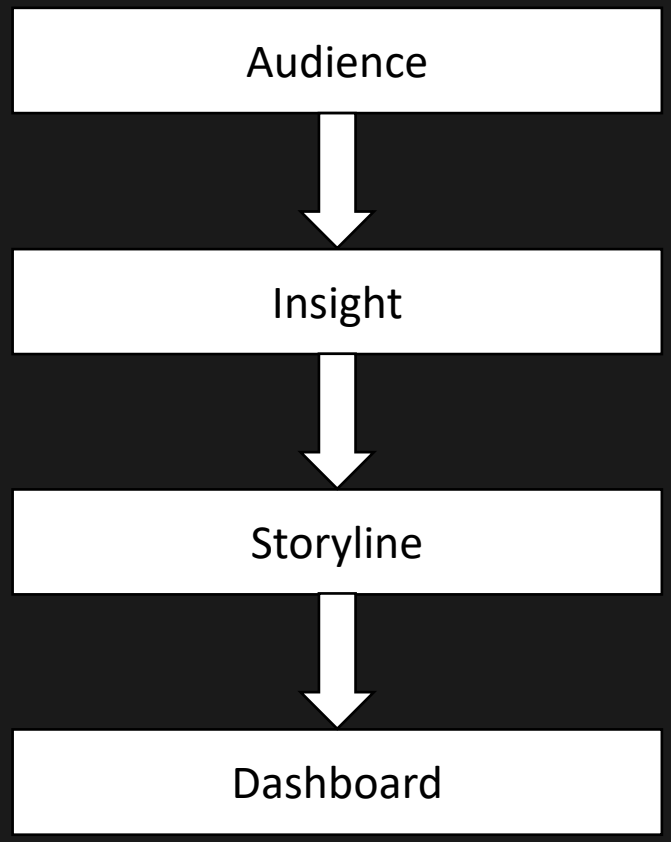
Purwadhika Digital Technology School | Digital For Executive

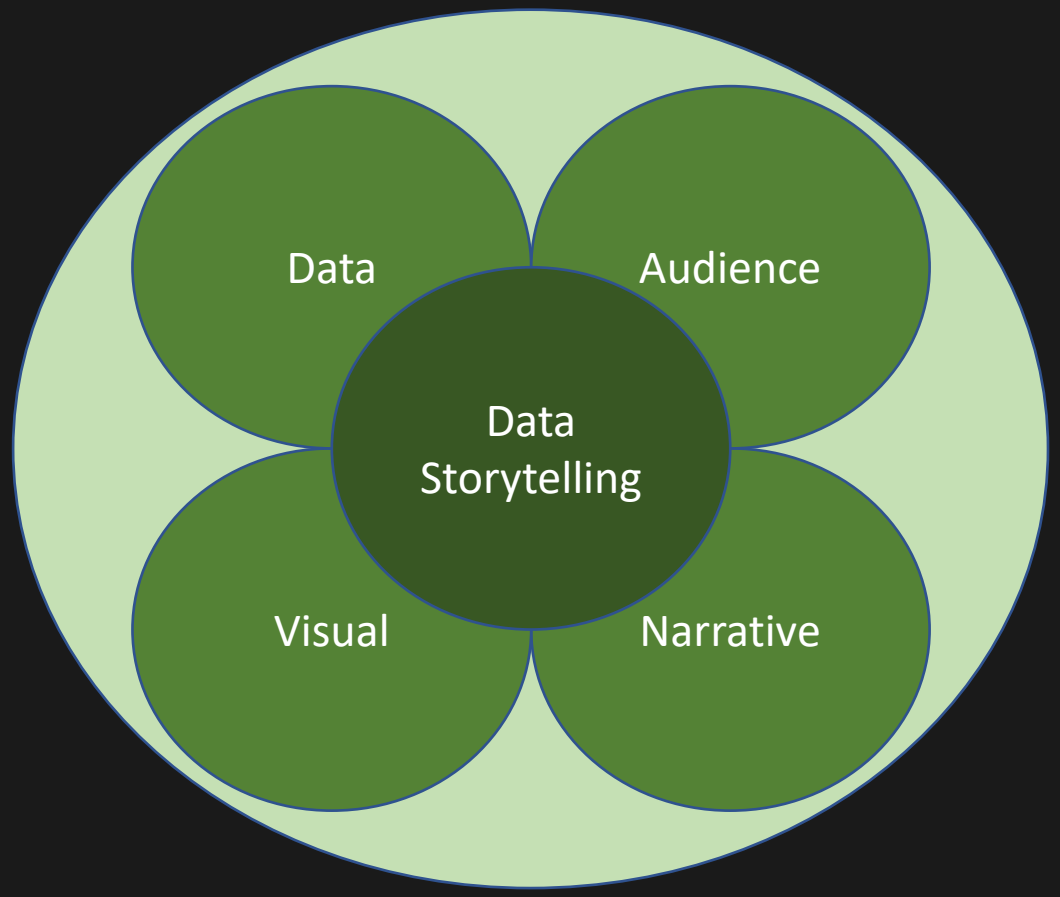
Data Analysis

Session 12:
Data Storytelling & Visualization

Digital for Executive

Data Storytelling




Data
Audience
Data Storytelling
Visual
Narrative

What You Should Know about Data Storytelling ?

- Stories are memorable and viral
- Deliver your messages not your work
- Better decision making
- Your tools don't know exactly which one is the story
- Exploratory vs explanatory analysis

Audience

- communicate to too many different people = hard position
- Narrow your target audience
  - **Driver**, who set the pace and pushing this forward
  - **Approver**, who decide the output is acceptabel
  - **Contributor**, who contribute to the process or output
  - **Informed**, Who needs awareness of the activity and output
- The more you know about audience the better communication

Align Your Messages with Audience Needs

Situation | Problems | Consequences | Action | Impact

Frame these components:
"As a **[persona]**, I'm in **[situation]** where I face **[problem]**, leading to **[consequence]**. Solving it by **[action]** leads to **[impact]**"

John: As a **Marketing manager**, I have to create **region-wise budget** for the next quarter. I don't know **which regions** give the **highest ROI**, so **my spend isn't optimized**. Solving it by prioritizing **the region** will lead to **maximum ROI**.

Insight

BIG

The analysis should be **statistically significant** and **numerically significant**.

SUPRISING

Is this something they didn't know ? Like **too obvious ?** Does it divide people opinion

USEFUL

You transform your data into **actionable insight**, Audience get better input about what should they do next.

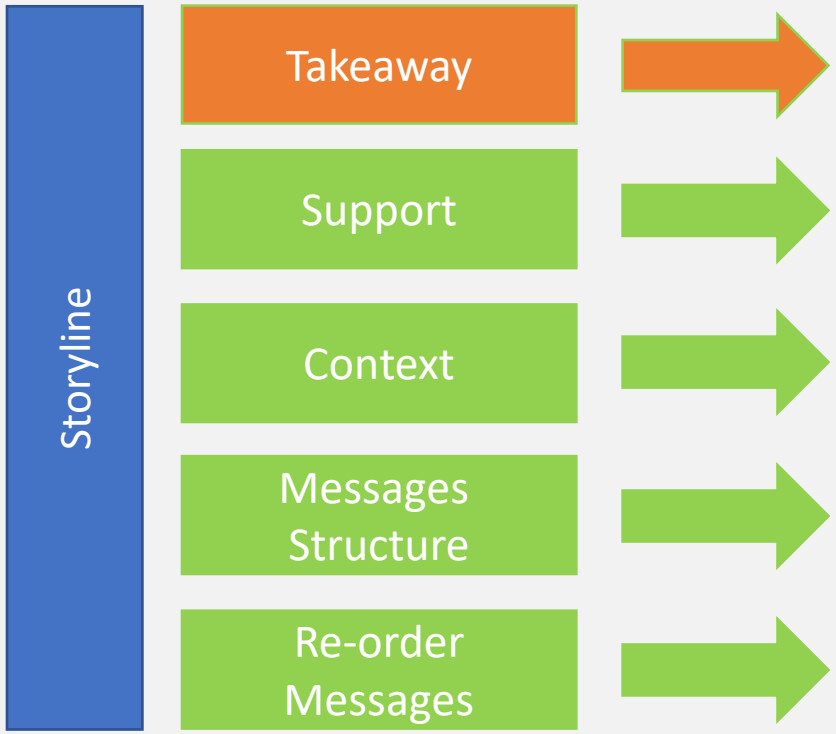
Insight Example

Insights	Big	Useful	Surprising
Twice as many Detractors talk about our Product's ease of use.	Low	Medium	High
Typing with capitalization in a credit application indicates creditworthiness	Low	Low	High
Almost 20% of all voice search queries are triggered by just 25 words	Low	High	Medium
More engaged employees have fewer accidents	Low	High	Low
About 50% of American small businesses do not have a website	High	Medium	Low
The recommendation system influences about 80% of content streamed on Netflix	Big	Low	Low

At least medium for all aspect (big, useful, surprising)

Storyline

- **Takeaway** → What is your main messages that audience should take away ?
- **Support** → Find analysis that supports your takeaway. Ignore irrelevant content
- **Context** → Convert analysis into messages by adding context
- **Messages Structure** → Structure the messages into a pyramid or a tree
- **Re-order Messages** → Re-order the messages to increase memorability and motivation



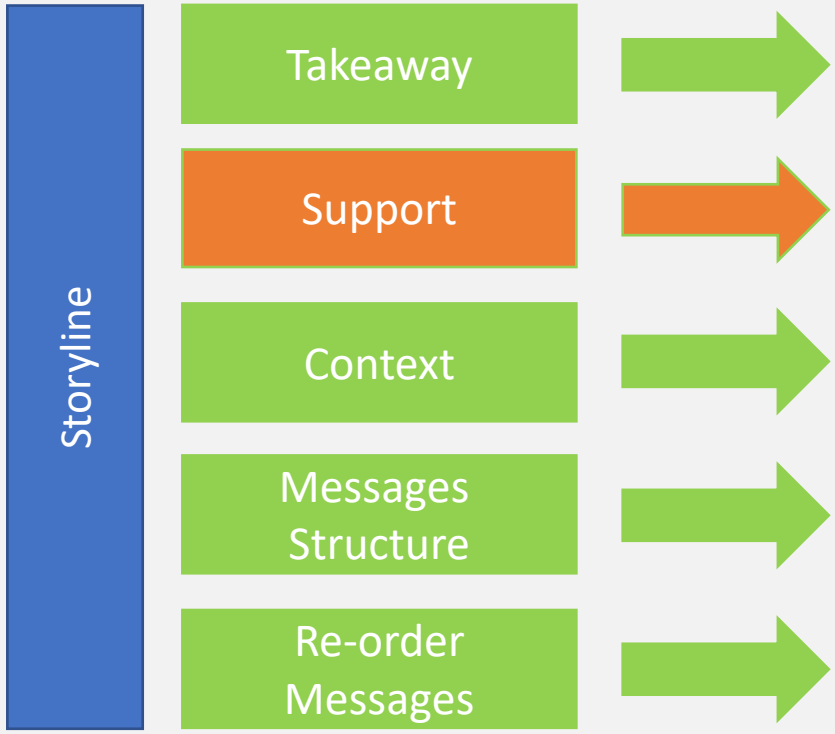
Takeaway is one-sentence summary of the most important messages for the audience

Analogy:

- moral story of a childhood tale
- can you easily summarize the business presentation in one sentence

Verify:

- Is your takeaway one single complete sentence ?
- Does your takeaway deliver what audience needs
- Will your audience care about that ?



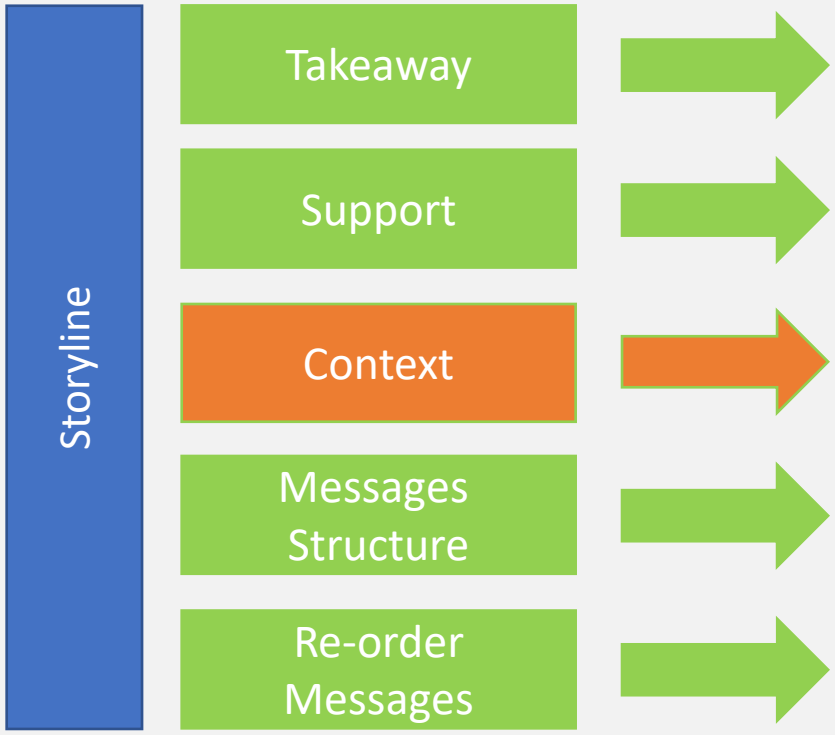
Only include analysis that proves the takeaway. Ensure that they fully prove the takeaway

How to ?

- List all possible analysis that support your takeaway
- Re word them as sentences
- Strike off what's not relevant

Verify :

- Is each your analysis support your take away
- Do the analysis prove the takeaway



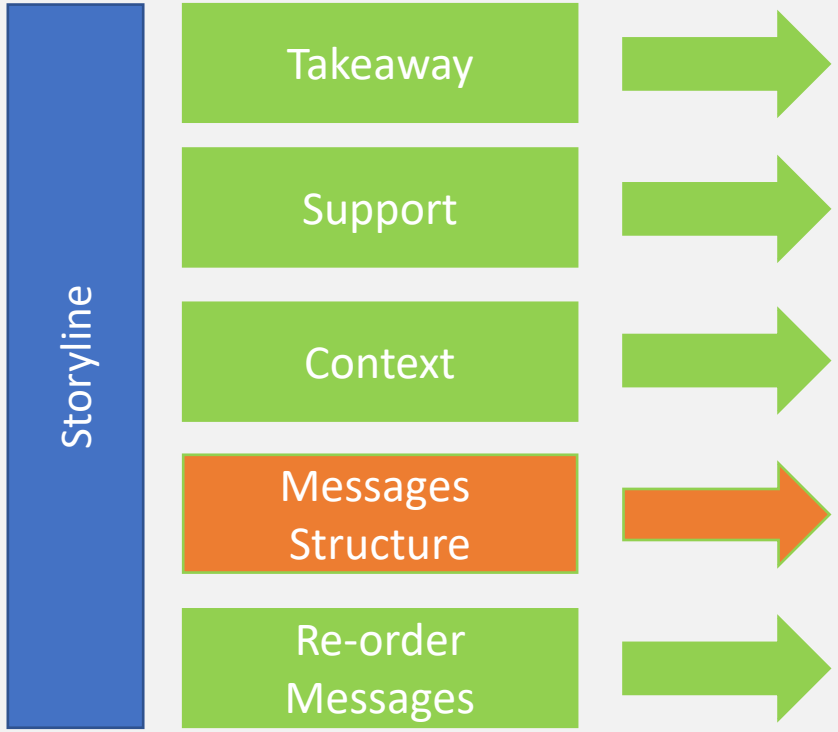
Frame each analysis as a message that the audience will understand and find relevant

Take your relevant analysis and add some context
How to add context ?

- Compare :Our current sales is $2.1 million lower than our past sales
- Analogies :If we deploy this marketing method, it will increase our revenue to $4 million
- etc

Verify :

- Will your audience easily understand your analysis ?
- Will they understand why this messages is relevant ?



Arrange messages hierarchically to prove & support the parent message

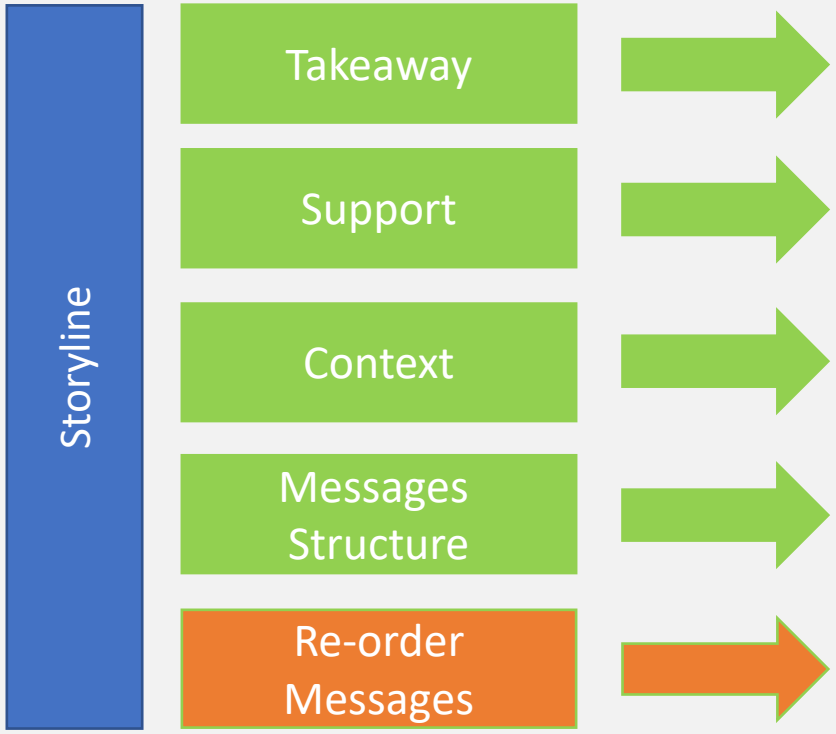
How to ?

- Start with take away
- Add a messages that support the take away
- Add next messages as required
- Add detail for each supporting messages

Example :

Launch sales were 30% less than target due to high competition

- Launch sales were projected at $20 mn in the first month, but achieved only $14 mn
  - Sales in every region were 20-50% lower.
  - ~~Only Philippines & Korea were on target~~
- Competitors discounted price by 35% - which is unsustainable for them
  - 80 store discounts increased from 15% to 35%
  - The maximum sustainable discount is 20%
- Stores offered higher discounts saw less than 20% of our target sales



Re-order the messages to increase memorability and motivation.

How to ?
Memorable flow:

1. Time : Present, past, future
2. Place : Jakarta, Bekasi, Bogor
3. Aspect : company, competitor, etc
4. Benefits : better faster or cheaper
5. Scale : local
6. Balance : pros, cons
7. Priority : order least to most important

or

Motivating sequences:
Order messages into an emotionally contrasting, motivating sequence.

Data Visualization

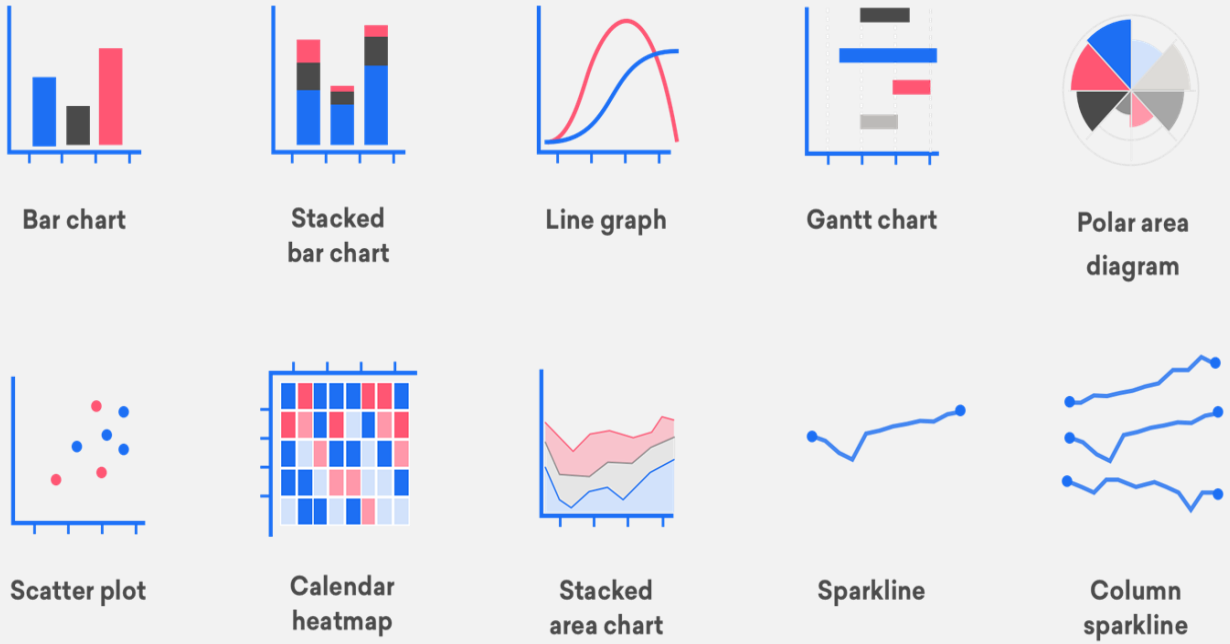
Outline

- Basic Concept of Visualization
- Type of Visualization
- Tools for visualization
- Commonly used chart

Objective

- Understand The Concept of Creating data visualization
- Answering analytics question using data visualization

What is Data Visualization ?



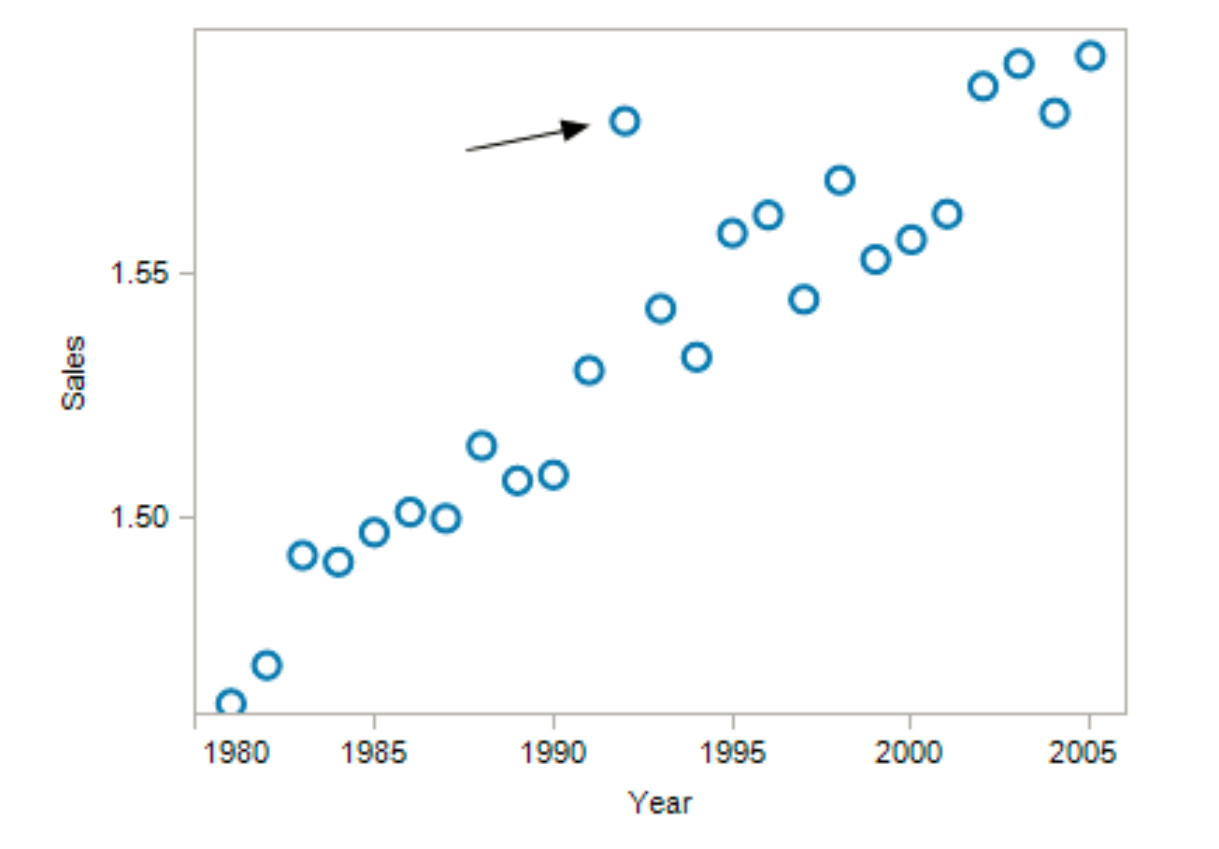
- Data Visualization is the presentation of data in pictorial or graphical format
- Data Visualization is a discipline to understand data by serving it visually so any **pattern or trends**, **composition**, **comparison** and **relationship** can be exposed

Why is Data Visualization very important ?

- You can utilize data visualization to **explore your data**. Exploring your data can improve your chance to find insight.
- You can also **share your findings** with other by using data visualization. Human brain is easier to process any information using visualization than spreadsheet or report
- Quick and easy way to convey concept in a universal manner.

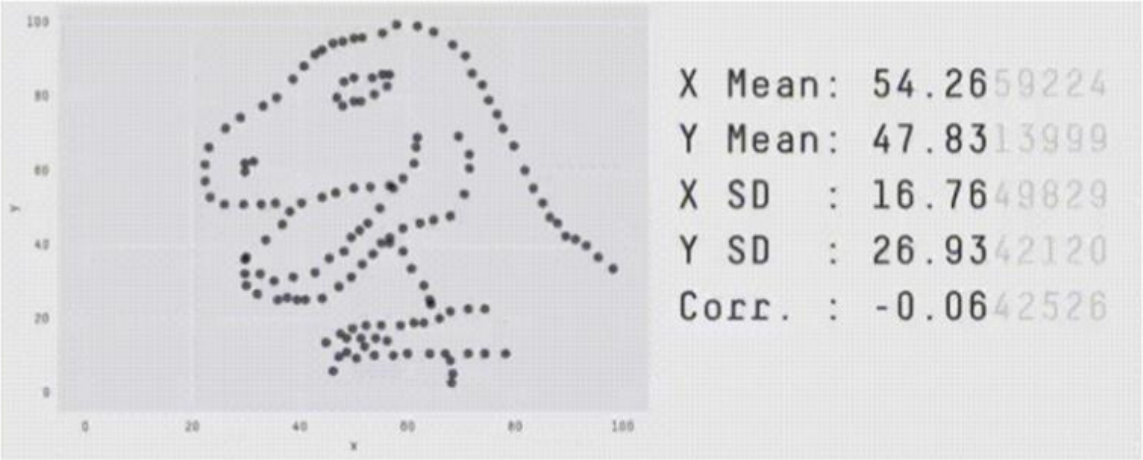
Which method is easier to identify uncommon event (anomaly) between year vs sales ?

	A	B
1	Year	Sales
2	1981	1.4622
3	1982	1.47004
4	1983	1.49253
5	1984	1.49118
6	1985	1.49722
7	1986	1.50138
8	1987	1.50008
9	1988	1.51493
10	1989	1.50781
11	1990	1.50899
12	1991	1.53037
13	1992	1.58137
14	1993	1.54299
15	1994	1.53307
16	1995	1.55845
17	1996	1.56213
18	1997	1.54488
19	1998	1.56927
20	1999	1.55305
21	2000	1.5571
22	2001	1.56235
23	2002	1.58847
24	2003	1.59309
25	2004	1.58303
26	2005	1.5947



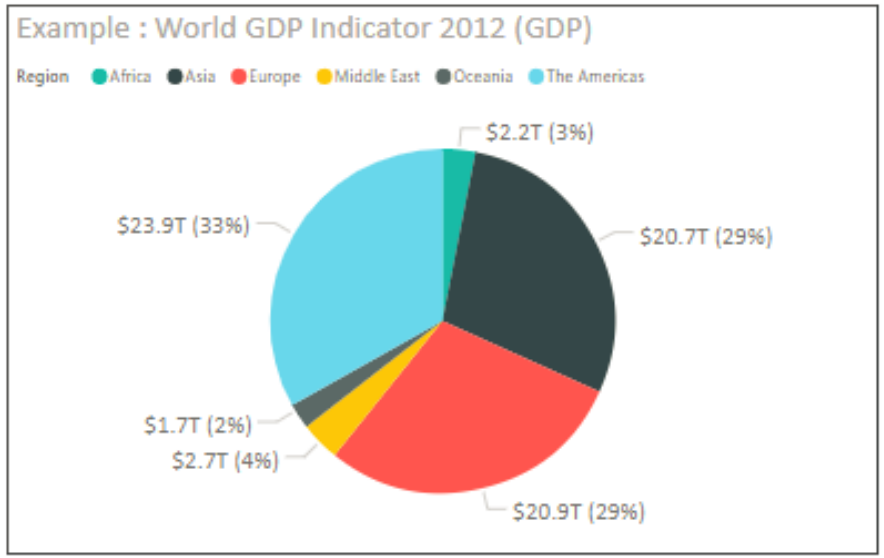
Visualization can be more meaningful

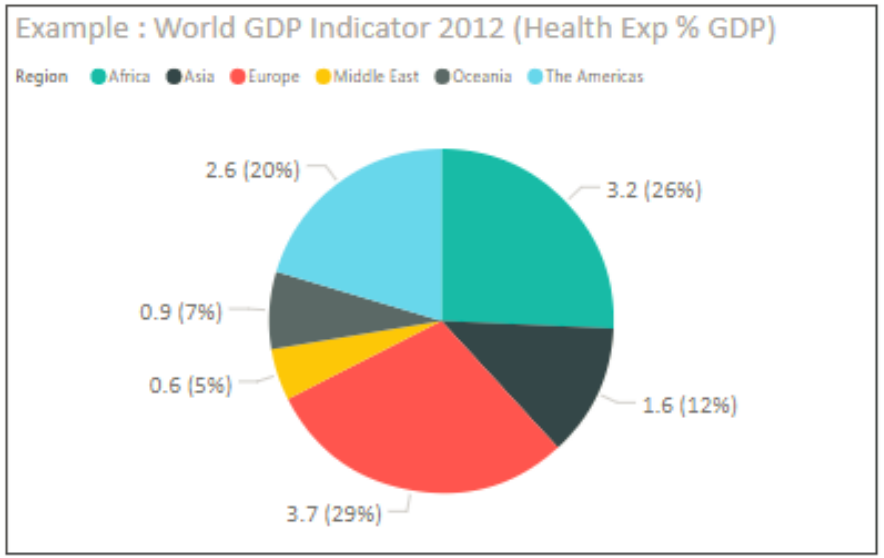
- It's hard to find patterns & derive insights from raw data
- Statistics can summarize data, but may hide patterns in how the data is spread
- We use visual encoding techniques to map data to visual attributes



Example : World GDP Indicator 2012

Region	Population 0-14	Population 15-64	Population 65+	Population Urban	GDP	Health Exp % GDP	Health Exp/Capita
Africa	21.30	29.84	1.86	22.04	$2,232,736,028,370	3.24	$7,941
Asia	9.01	22.66	2.33	17.10	$20,689,754,411,401	1.57	$14,214
Europe	6.72	27.84	6.44	32.79	$20,930,307,966,112	3.69	$120,452
Middle East	3.50	9.01	0.49	10.22	$2,681,972,907,100	0.61	$11,424
Oceania	4.00	8.13	0.87	7.75	$1,727,175,612,930	0.92	$11,652
The Americas	9.88	24.11	3.01	29.48	$23,926,023,856,769	2.57	$33,654
Total	**54.41**	**121.58**	**15.00**	**119.38**	**$72,187,970,782,682**	**12.59**	**$199,337**





QUESTION:
What region has the highest and lowest GDP and Health Exp % GDP?

Source data : World Indicator (Power BI)

Some questions that can be answered through data visualization

How is the trend of the sales this past 3 months?

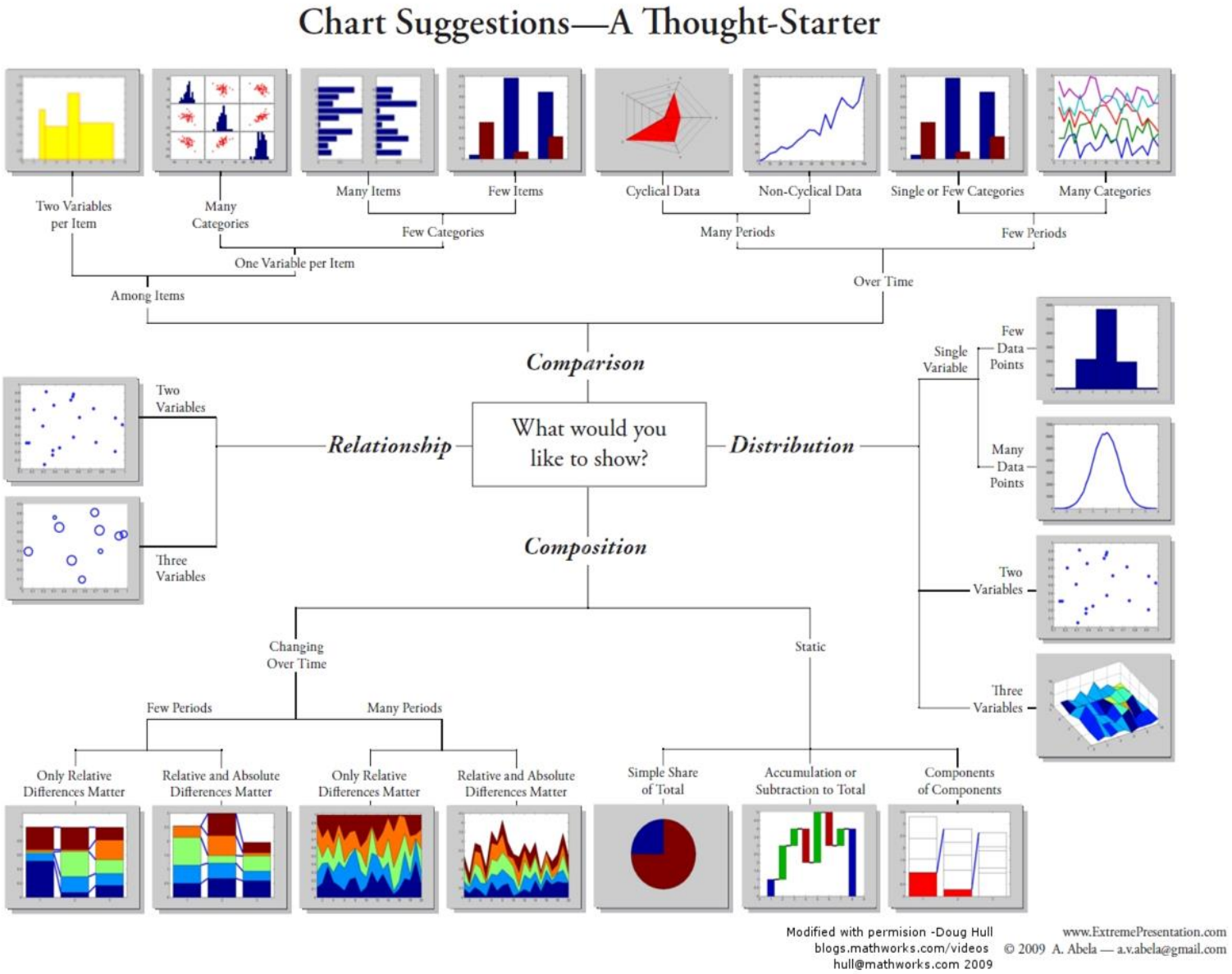
Which Factor influence customer behavior?

How to place your product to improve sales?

Which products has the highest sales?

Data Visualization Types:

1. Comparison
2. Relationship
3. Distribution
4. Composition



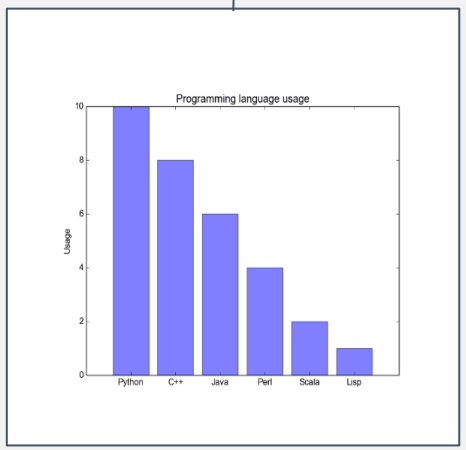
Data Visualization Type

- Comparison
- Relationship
- Distribution
- Composition

Comparison Among Items

One variables per items

Few items



Barchart or barplot

Many items

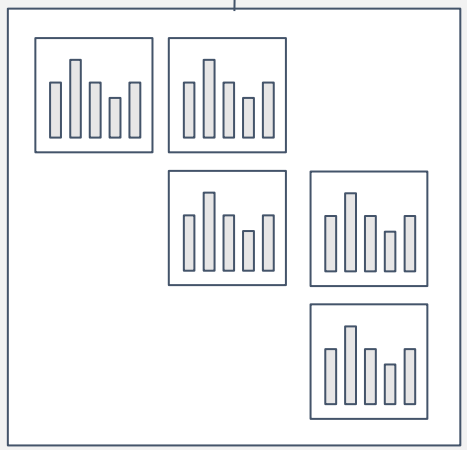
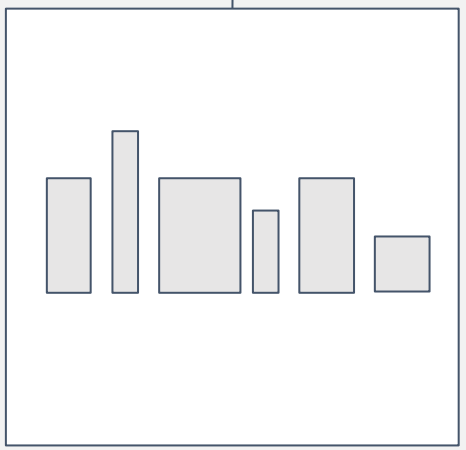


Table with embedded chart

Two variables per items



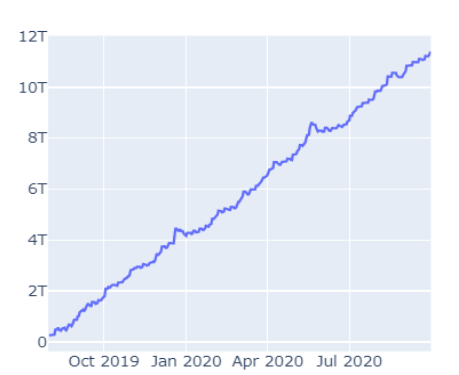
Variable with column chart

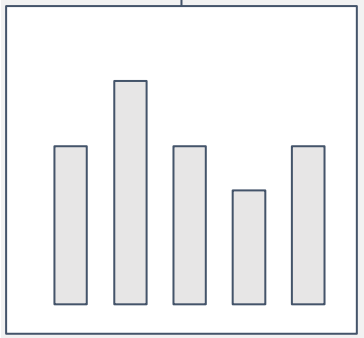
Data Visualization Type

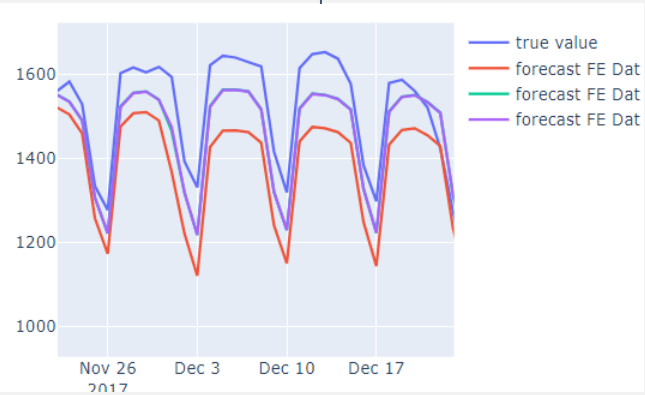
- Comparison
- Relationship
- Distribution
- Composition

Comparison Over Times

- One variables
  - Many Periods
    - Linechart
  - Few Periods
    - Barchart or barplot
- Many variables / Same variables different categories
  - Multiple Line chart





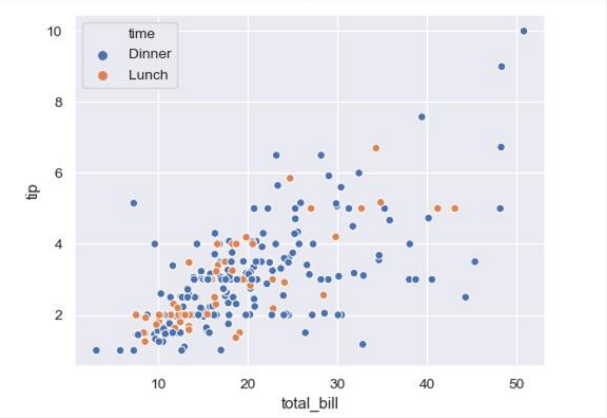


Data Visualization Type

- Comparison
- Relationship
- Distribution
- Composition

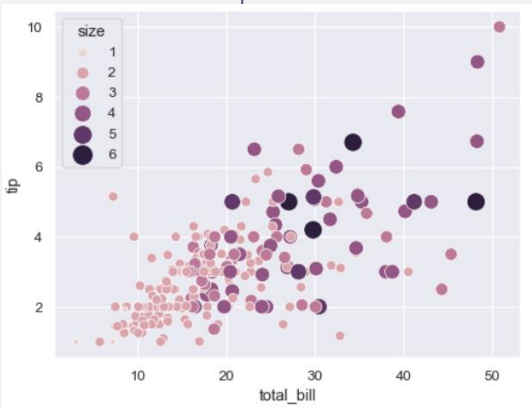
Relationship

Two variables



Scatterplot

Three variables



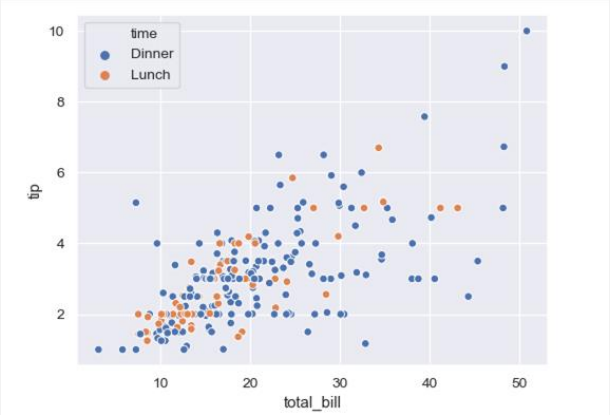
Scatterplot with Buble

Data Visualization Type

- Comparison
- Relationship
- Distribution
- Composition

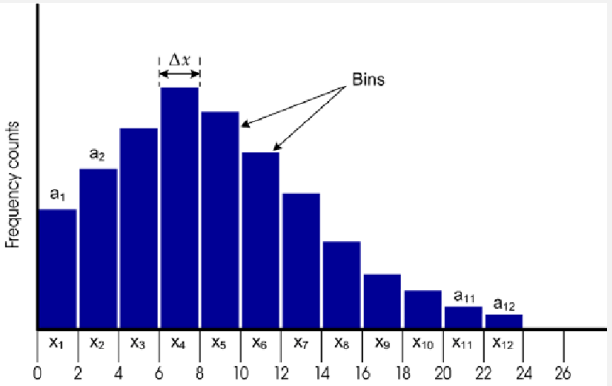
Distribution

Two variables



Scatterplot

One Variable



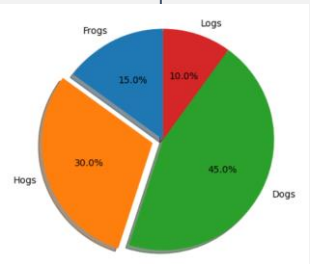
Histogram

Data Visualization Type

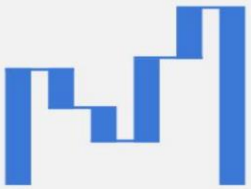
- Comparison
- Relationship
- Distribution
- Composition

Composition Static

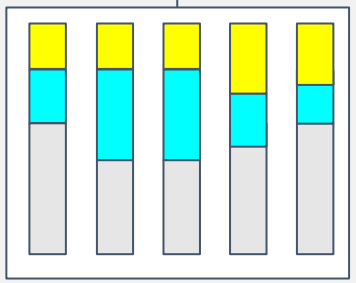
- Share of total
- Accumulation or subtraction of total
- Share of total per items or category
- Share of total many items or category



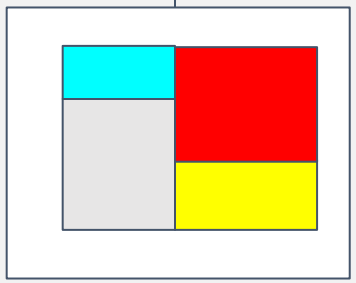
Pie chart



Waterfall chart



Stacked barchart (100%)



Tree map

Data Visualization Type

- Comparison
- Relationship
- Distribution
- Composition

Composition Overtime

- Few Periods
  - Relative difference
    - Stacked barchart 100%
  - Absolute difference
    - Stacked barchart
- Many Periods
  - Relative difference
    - Stacked area chart 100%
  - Absolute difference
    - Stacked area chart

Tool for Visualization

Python:

- Pandas
- Matplotlib
- Seaborn
- Plotly
- Bokeh
- Etc

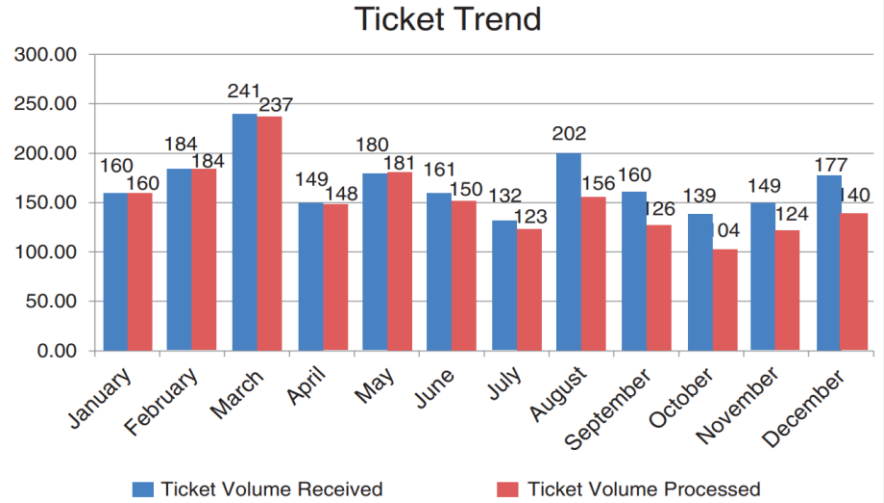
Commonly used:

- *Power BI*
- **Tableau**
- Looker
- Qlik
- Etc

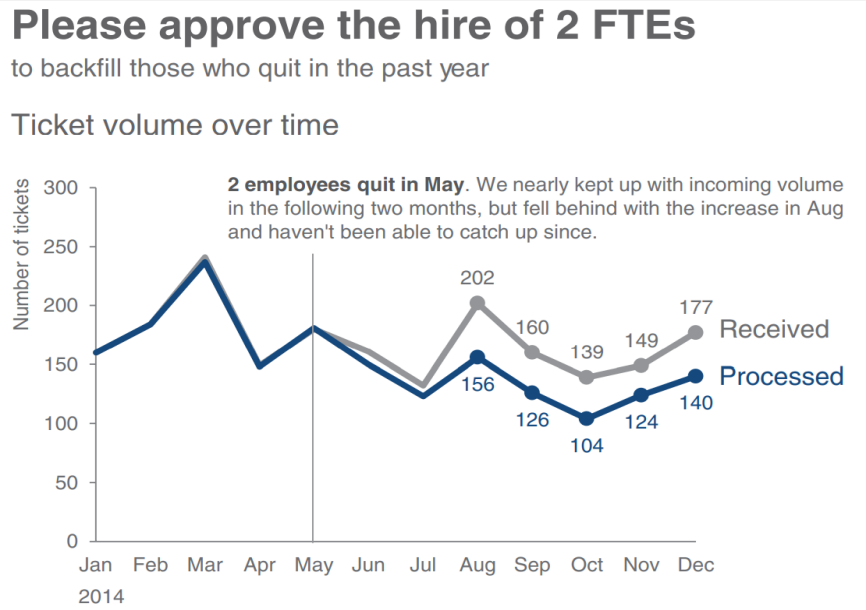
How to Generate Data Visualization ?

$$\text{Data Ink Ratio} = \frac{\text{Data Ink Ratio}}{\text{Totak Ink for all graphic}}$$

- Add title to your data visualization to make it more informative
- Label your axis (x-axis and w-axis)
- Label your graph (legend)
- Mark interesting data points
- Play with color and size



Bad Ink Ratio



Better Ink Ratio

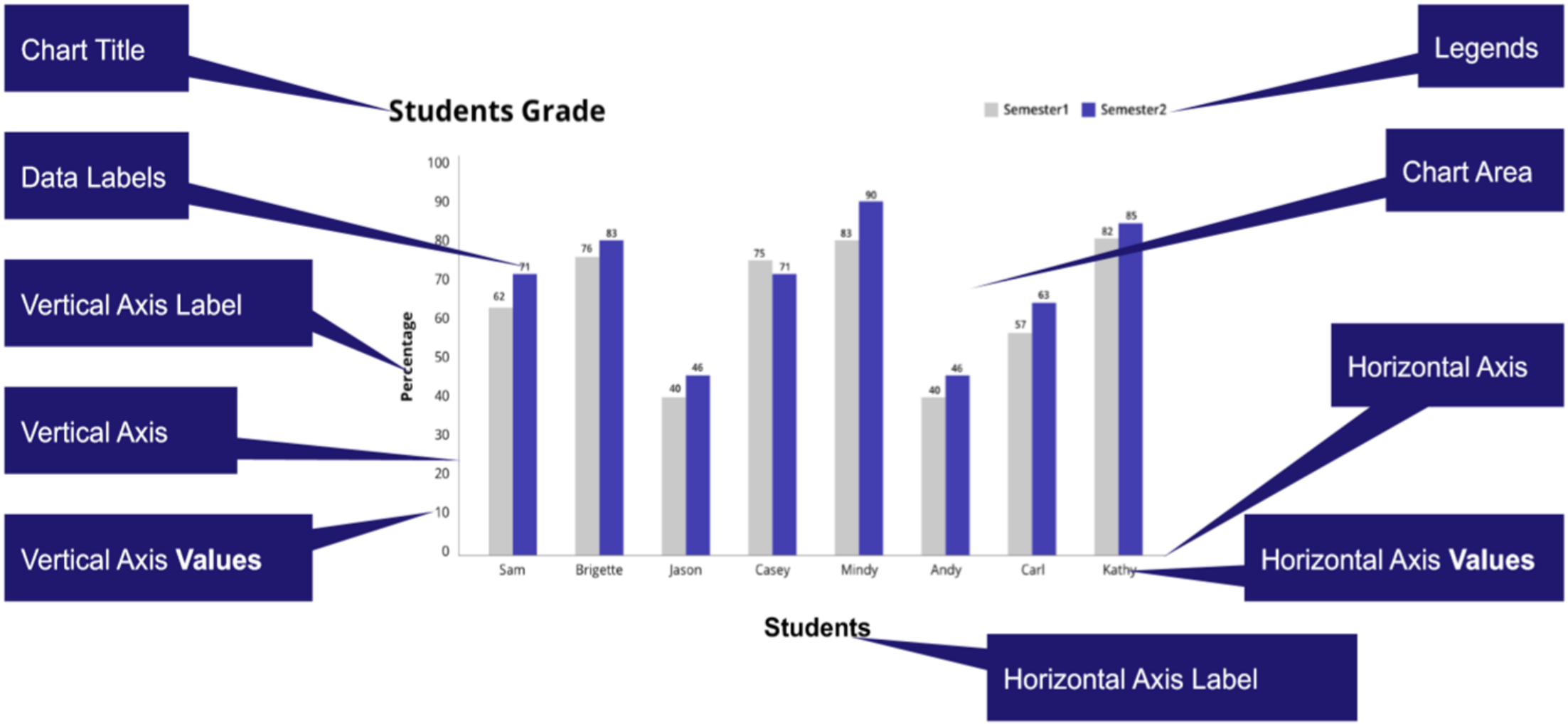
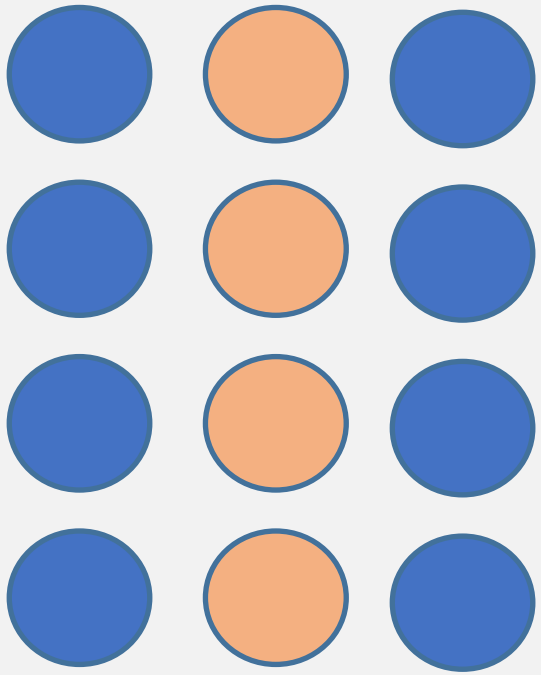
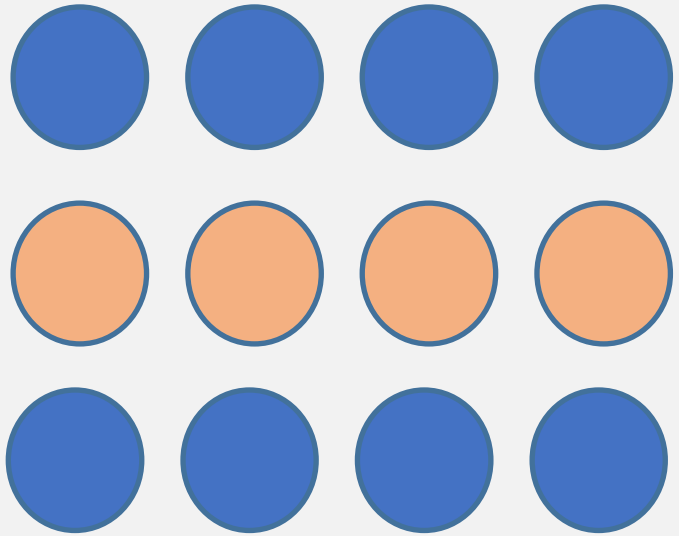


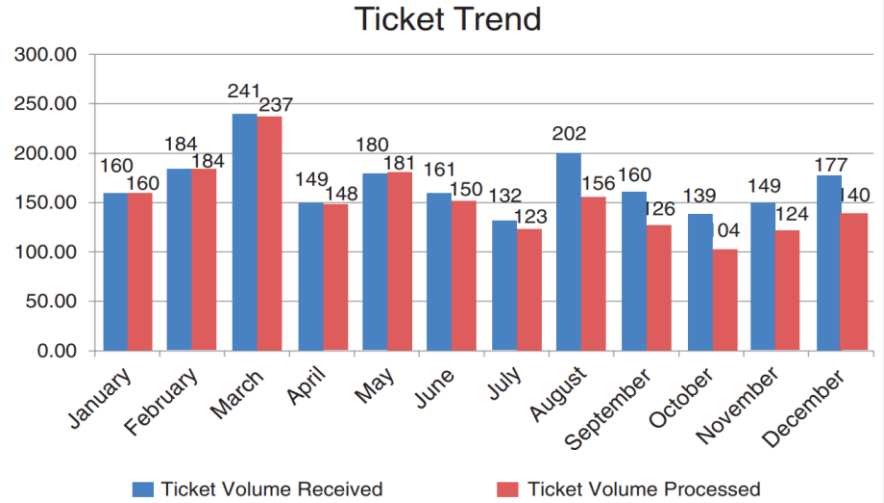
Chart Title
Data Labels
Vertical Axis Label
Vertical Axis
Vertical Axis Values
Legends
Chart Area
Horizontal Axis
Horizontal Axis Values
Horizontal Axis Label
Students Grade
Semester1
Semester2
Percentage
100
90
80
70
60
50
40
30
20
10
0
62
71
76
83
40
46
75
71
83
90
40
46
57
63
82
85
Sam
Brigette
Jason
Casey
Mindy
Andy
Carl
Kathy
Students

The Gstat Law of Similarity

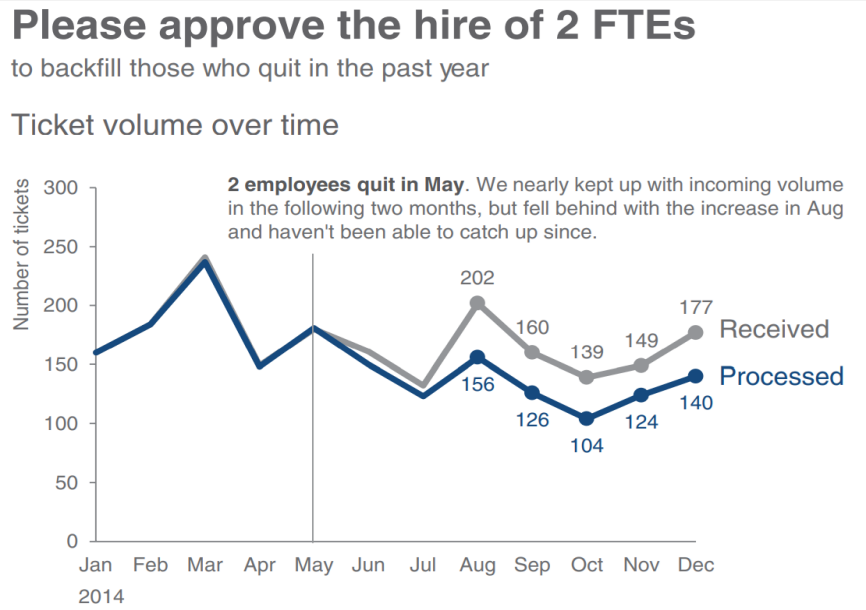


Preferable Approach





Bad Legend Positioning



Better Legend Positioning

Thank You!

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